# MODULE 1: CHEMICAL THERMODYNAMICS AND KINETICS

**Official FAT Study Textbook & Step-by-Step Question Bank Solutions (CO: 1)**

Aligned with Official VIT BACHY101 Syllabus, Department Question Bank, and Past FAT Papers

**Scope of Module 1 (Official VIT Syllabus - All 9 Topics):**

(1) Introduction terminologies, (2) Zeroth law - First law of thermodynamics, Enthalpy and heat capacity, (3) Applications of first law - Second law of thermodynamics, (4) Carnot cycle, (5) Third law of thermodynamics, Gibbs-Helmholtz equation, (6) Chemical kinetics: Rate equation, Order of reaction, Integral rate equations, Half-life time, (7) Collision theory of reaction rate, Effect of temperature and catalysts, (8) Worked Numericals - Part 1, and (9) Worked Numericals - Part 2.

## 1. Core Syllabus Definitions & Theoretical Foundations (Line-by-Line)

**1. Open System**

A thermodynamic system bounded by a permeable interface that freely exchanges both matter and energy with its surroundings (e.g. an open beaker containing boiling water).

**2. Closed System**

A thermodynamic system that exchanges energy in the form of heat and boundary work with its surroundings but cannot exchange mass across its boundaries (e.g. gas enclosed in a cylinder with a tightly sealed movable piston).

**3. Isolated System**

A thermodynamic system completely bounded by rigid, adiabatic, impermeable walls that exchanges neither matter nor energy with its surroundings (e.g. hot liquid enclosed in an ideal vacuum dewar flask).

**4. Intensive Properties**

Physical or thermodynamic properties whose values are independent of the total quantity or mass of matter present in the system (e.g. Temperature $T$, Pressure $P$, Density $\rho$, Refractive index, Boiling point, Molar heat capacity $C_m$).

**5. Extensive Properties**

Thermodynamic properties whose values scale directly in proportion to the total quantity or mass of the system (e.g. Volume $V$, Mass $m$, Internal energy $U$, Enthalpy $H$, Entropy $S$, Total heat capacity $C$).

**6. State Functions**

Thermodynamic equilibrium properties whose values depend exclusively on the present equilibrium

state of the system, irrespective of the historical path or mechanism taken to reach that state (e.g. $P, V, T, U, H, S, G$).

**7. Path Functions**
Thermodynamic quantities whose magnitudes depend directly on the specific transformation pathway followed during the thermodynamic transition (e.g. Heat absorbed $q$, Boundary work done $w$).

**8. Zeroth Law of Thermodynamics**
If body A is in thermal equilibrium with body B, and body B is simultaneously in thermal equilibrium with body C, then body A and body C are in mutual thermal equilibrium. This provides the absolute scientific foundation for temperature measurement.

**9. First Law of Thermodynamics**
The principle of conservation of energy: Energy cannot be created or destroyed, only transformed between equivalent forms:

$$\Delta U = q + w$$

where $\Delta U$ is internal energy change, $q$ is heat absorbed by the system, and $w$ is work performed on the system ($w = -P_{\text{ext}}\Delta V$). For free expansion into a vacuum ($P_{\text{ext}} = 0$), work done is strictly zero ($w_{\text{vacuum}} = 0$).

**10. Enthalpy & Mayer's Relation**
Enthalpy represents total thermal heat content defined as $H = U + PV$. At constant pressure $q_p = \Delta H$; at constant volume $q_v = \Delta U$. For one mole of an ideal gas ($PV = RT$), differentiating $H = U + RT$ with respect to temperature at constant pressure yields Mayer's proof:

$$C_p - C_v = R$$

**11. Reversible Adiabatic Expansion**
A thermodynamic process with zero heat exchange with the surroundings ($dq = 0$). The system performs boundary expansion work strictly at the expense of its own internal energy ($\Delta U = w < 0$), causing the temperature to drop:

$$PV^{\gamma} = \text{constant}, \quad TV^{\gamma-1} = \text{constant}$$

Sudden discharge from a high-pressure $CO_2$ fire extinguisher produces rapid adiabatic expansion and cooling, freezing into solid dry ice snow.

**12. Second Law of Thermodynamics**
Natural spontaneous processes occur with an increase in total entropy of the universe. Clausius statement: Heat cannot spontaneously flow from a colder reservoir to a hotter reservoir without external work. Kelvin-Planck statement: No cyclic heat engine can absorb heat from a single thermal reservoir and convert it completely into equivalent mechanical work without waste heat rejected to a sink.

**13. Carnot Cycle & Thermal Efficiency**
An idealized four-stroke reversible cyclic heat engine operating between hot reservoir $T_H$ and cold

sink $T_C$ (Isothermal expansion → Adiabatic expansion → Isothermal compression → Adiabatic compression). The maximum theoretical thermal efficiency is:

$$\eta = \frac{T_H - T_C}{T_H} = 1 - \left(\frac{T_C}{T_H}\right)$$

### 14. Third Law of Thermodynamics

The entropy of a pure, perfectly crystalline substance at absolute zero ($T = 0$ K) is exactly zero:

$$\lim_{T \to 0} S = 0$$

This provides an absolute reference datum for evaluating standard molar entropies $S^\circ$.

### 15. Gibbs Free Energy & Reaction Spontaneity

A thermodynamic state function defined as $G = H - TS$. Governed at constant temperature and pressure by the Gibbs-Helmholtz relation:

$$\Delta G = \Delta H - T\Delta S$$

- $\Delta G < 0$: Spontaneous process (thermodynamically favored).
- $\Delta G = 0$: Dynamic equilibrium state (crossover temperature $T_{\text{eq}} = \Delta \frac{H}{\Delta} S$).
- $\Delta G > 0$: Non-spontaneous process (requires external driving force).

### 16. Rate of Chemical Reaction

The instantaneous rate of change in concentration of reactants or products per unit time: $\text{Rate} = -\frac{d[A]}{d}t = k[A]^n$, where $k$ is the specific rate constant and $n$ is reaction order.

### 17. Zero-Order Kinetics

The reaction rate is completely independent of reactant concentration ($\text{Rate} = k$). Integrated rate law: $[A]_t = [A]_0 - kt$. Half-life: $t_{\frac{1}{2}} = \frac{[A]_0}{2k}$.

### 18. First-Order Kinetics

The reaction rate is directly proportional to the first power of reactant concentration. Integrated rate law: $\ln\left(\frac{[A]_0}{[A]_t}\right) = kt$ or $k = \left(\frac{2.303}{t}\right) \log_{10}\left(\frac{[A]_0}{[A]_t}\right)$. The half-life is strictly constant and independent of initial concentration: $t_{\frac{1}{2}} = \frac{0.693}{k}$.

### 19. Second-Order Kinetics

The reaction rate depends on the square of one reactant concentration or the product of two concentrations. Integrated rate law: $\frac{1}{[A]_t} = \frac{1}{[A]_0} + kt$. The half-life is inversely proportional to initial concentration: $t_{\frac{1}{2}} = \frac{1}{k[A]_0}$.

### 20. Arrhenius Equation & Activation Energy

Governs the exponential dependence of reaction rate constant $k$ on absolute temperature $T$:

$$k = Ae^{-\frac{E_a}{RT}}$$

where $A$ is the pre-exponential frequency factor, $R = 8.314$ J/(mol · K), and $E_a$ is the threshold activation energy. Comparing rate constants at two temperatures:

$$\ln\left(\frac{k_2}{k_1}\right) = \left(\frac{E_a}{R}\right)\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

# 2. In-Depth Chemical Theory & Step-by-Step Derivations

## Derivation of Mayer's Relation ($C_p - C_v = R$)

For one mole of an ideal gas, enthalpy is defined as:

$$H = U + PV = U + RT$$

Differentiating with respect to temperature $T$ at constant pressure:

$$\left(\partial \frac{H}{\partial} T\right)_P = \left(\partial \frac{U}{\partial} T\right)_P + R$$

Since internal energy $U$ of an ideal gas depends solely on temperature: $\left(\partial \frac{U}{\partial} T\right)_P = \left(\partial \frac{U}{\partial} T\right)_V = C_v$. By definition, $\left(\partial \frac{H}{\partial} T\right)_P = C_p$. Substituting yields the celebrated relation:

$$C_p = C_v + R \Rightarrow \boldsymbol{C_p - C_v = R}$$

## The 4-Stroke Carnot Cycle & Efficiency Derivation

A Carnot heat engine operates between $T_H$ and $T_C$ through four reversible stages:

1. **Stroke 1 (Isothermal Expansion, A → B at $T_H$):** Gas absorbs heat $q_H$ from reservoir: $q_H = -w_1 = RT_H \ln\left(\frac{V_B}{V_A}\right)$. Internal energy change $\Delta U_1 = 0$.
2. **Stroke 2 (Adiabatic Expansion, B → C from $T_H$ to $T_C$):** Thermally insulated ($q = 0$). Gas expands doing work: $w_2 = C_v(T_C - T_H) = -C_v(T_H - T_C)$.
3. **Stroke 3 (Isothermal Compression, C → D at $T_C$):** Heat $q_C$ rejected to cold sink: $q_C = -w_3 = RT_C \ln\left(\frac{V_D}{V_C}\right) = -RT_C \ln\left(\frac{V_C}{V_D}\right)$.
4. **Stroke 4 (Adiabatic Compression, D → A from $T_C$ to $T_H$):** Thermally insulated ($q = 0$). Work performed on gas: $w_4 = C_v(T_H - T_C)$.

Summing work over the entire cycle:

$$w_{\text{net}} = w_1 + w_2 + w_3 + w_4 = -RT_H \ln\left(\frac{V_B}{V_A}\right) - RT_C \ln\left(\frac{V_D}{V_C}\right)$$

Since paths BC and DA are adiabatic: $T_H V_B^{\gamma-1} = T_C V_C^{\gamma-1}$ and $T_H V_A^{\gamma-1} = T_C V_D^{\gamma-1}$, which gives $\frac{V_B}{V_A} = \frac{V_C}{V_D}$. Therefore:

$$|w_{\text{net}}| = R(T_H - T_C) \ln\left(\frac{V_B}{V_A}\right)$$

The thermal efficiency $\eta$ is the ratio of useful net work output to heat absorbed at the hot reservoir:

$$\eta = \frac{|w_{\text{net}}|}{q_H} = \frac{R(T_H - T_C)\ln\left(\frac{V_B}{V_A}\right)}{RT_H \ln\left(\frac{V_B}{V_A}\right)} = \frac{\boldsymbol{T_H - T_C}}{\boldsymbol{T_H}} = \mathbf{1} - \left(\frac{\boldsymbol{T_C}}{\boldsymbol{T_H}}\right)$$

# 3. VIT FAT Exam Blueprint & Master Formula Sheet

Module 1 carries **15 to 20 marks** in BACHY101 FAT:

- **Question Type A (10 Marks):** Carnot engine efficiency derivation, heat absorbed, net cyclic work area, and refrigerator COP.
- **Question Type B (5-10 Marks):** Mayer's relation proof ($C_p - C_v = R$) and reversible adiabatic expansions ($PV^\gamma = C$).

- **Question Type C (10 Marks):** Integrated rate equations for 0th, 1st, and 2nd order reactions, half-life problems, and Arrhenius activation energy calculations.

| Concept / Law | Governing Equation / Formula | VIT Exam Key Insight |
|---|---|---|
| First Law of Thermodynamics | $\Delta U = q + w, \quad w = -\int P_{\text{ext}} dV$ | Free expansion into vacuum: $P_{\text{ext}} = 0 \rightarrow w = 0$. |
| Heat Capacity Relation | $C_p - C_v = R = 8.314\ \text{J/(mol} \cdot \text{K)}$ | Applies strictly to ideal gases ($PV = nRT$). |
| Carnot Engine Efficiency | $\eta = \frac{T_H - T_C}{T_H} = 1 - \left(\frac{T_C}{T_H}\right)$ | Temperatures $T_H, T_C$ MUST be converted to Kelvin! |
| Gibbs Free Energy | $\Delta G = \Delta H - T\Delta S$ | Spontaneous if $\Delta G < 0$; Equilibrium at $T_{\text{eq}} = \Delta \frac{H}{\Delta} S$. |
| Gibbs-Helmholtz Equation | $\Delta G = \Delta H + T\left[\frac{\partial(\Delta G)}{\partial T}\right]_P$ | Alternatively: $\left[\frac{\partial\left(\Delta \frac{G}{T}\right)}{\partial T}\right]_P = -\Delta \frac{H}{T^2}$. |
| First-Order Rate Law | $k = \left(\frac{2.303}{t}\right) \log_{10}\left(\frac{a}{a-x}\right)$ | Half life $t_{\frac{1}{2}} = \frac{0.693}{k}$ is independent of initial conc $a$. |
| Second-Order Rate Law | $k = \left(\frac{1}{t}\right)\left(\frac{x}{a(a-x)}\right)$ | Half life $t_{\frac{1}{2}} = \frac{1}{ka}$ is inversely proportional to $a$. |
| Arrhenius Equation | $k = Ae^{-\frac{E_a}{RT}} \Rightarrow$ $\ln\left(\frac{k_2}{k_1}\right) = \left(\frac{E_a}{R}\right)\left[\frac{T_2 - T_1}{T_1 T_2}\right]$ | $R = 8.314\ \text{J/(mol} \cdot \text{K)}$; activation energy $E_a$ is in J/mol. |

## 4. Worked VIT Question Bank & Past FAT Numerical Problems

**VIT FAT Problem 1 (10 Marks): Carnot Engine Calculations**

A reversible Carnot heat engine operates between 500 ° C and 100 ° C. In each cycle, it absorbs 6000 J of heat from the higher-temperature source.
(a) Calculate the maximum thermal efficiency of this Carnot engine.
(b) Calculate the net useful mechanical work produced per cycle.
(c) Calculate the quantity of waste heat rejected to the cold sink per cycle.

**Step-by-Step Solution:**

- **Step 1: Absolute Temperature Conversion (Mandatory):**

$$T_H = 500 + 273.15 = 773.15\ \text{K}$$

$$T_C = 100 + 273.15 = 373.15\ \text{K}$$

- **Step 2: Maximum Carnot Efficiency:**

$$\eta = \frac{T_H - T_C}{T_H} = \frac{773.15 - 373.15}{773.15} = \frac{400}{773.15} = \mathbf{0.5174} \quad (\mathbf{51.74\%})$$

- **Step 3: Net Work Done ($w_{\text{net}}$):**

$$|w_{\text{net}}| = \eta \cdot q_H = 0.51736 \times 6000 \text{ J} = \mathbf{3104.2 \text{ J}}$$

- **Step 4: Heat Rejected to Cold Sink ($q_C$):**

$$|q_C| = q_H - |w_{\text{net}}| = 6000 - 3104.2 = \mathbf{2895.8 \text{ J}}$$

**Boxed Final Answers:**
(a) Efficiency $\eta = \mathbf{51.74\%}$
(b) Net Mechanical Work $w_{\text{net}} = \mathbf{3104.2 \text{ J}}$
(c) Waste Heat Rejected $q_C = \mathbf{2895.8 \text{ J}}$

**VIT FAT Problem 2 (10 Marks): First-Order Kinetics & Arrhenius Activation Energy**

(a) In the first-order decomposition of $C_4H_9Cl$, the concentration drops from 0.100 M to 0.050 M in 340 s. Calculate the rate constant $k$.
(b) A reaction rate quadruples ($\frac{k_2}{k_1} = 4.0$) when the temperature increases from 300 K to 320 K. Calculate the activation energy $E_a$ of the reaction in kJ/mol.

**Step-by-Step Solution:**

- **Part (a): Rate Constant Calculation:** The concentration drop from 0.100 M to 0.050 M represents exactly one half-life ($t_{\frac{1}{2}} = 340$ s):

$$k = \frac{\ln 2}{t_{\frac{1}{2}}} = \frac{0.69315}{340 \text{ s}} = \mathbf{2.039 \times 10^{-3} s^{-1}}$$

- **Part (b): Activation Energy from Arrhenius Ratio:**

$$\ln\left(\frac{k_2}{k_1}\right) = \left(\frac{E_a}{R}\right)\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

Substitute $\frac{k_2}{k_1} = 4$, $T_1 = 300$ K, $T_2 = 320$ K, and $R = 8.314$ J/(mol · K):

$$\ln(4.0) = 1.38629 = \left(\frac{E_a}{8.314}\right)\left[\frac{320 - 300}{300 \times 320}\right]$$

$$1.38629 = \left(\frac{E_a}{8.314}\right)\left[\frac{20}{96000}\right] = E_a \times (2.5061 \times 10^{-5})$$

$$E_a = \frac{1.38629}{2.5061 \times 10^{-5}} = 55317 \text{ J/mol} = \mathbf{55.32 \text{ kJ/mol}}$$

**Boxed Final Answers:**
(a) First-Order Rate Constant $k = \mathbf{2.04 \times 10^{-3} s^{-1}}$
(b) Activation Energy $E_a = \mathbf{55.32 \text{ kJ/mol}}$